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High-transferability black-box attack of binary image segmentation via adversarial example augmentation☆

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ABSTRACT

The application of deep neural networks (DNNs) has significantly advanced the binary image segmentation (BIS) task. However, DNNs have been found to be susceptible to adversarial attacks involving subtle perturbations. The existing black-box attack methods usually generate one single adversarial example for different target models, leading to poor transferability. To address this issue, this paper proposes a novel adversarial example augmentation (AEA) framework to improve the transferability of black-box attacks. Our method dedicates to generating an adversarial example set (AES) which contains a set of distinct adversarial examples. Specifically, we first employ an existing model as the surrogate model which is attacked to optimize the adversarial perturbation via maximizing the Binary Cross-Entropy (BCE) loss between the prediction of the surrogate model and the pseudo label, thus producing a sequence of adversarial examples. During the optimization process, besides the BCE loss, we additionally introduce deep feature losses among different adversarial examples to fully distinguish the generated adversarial examples. In this way, we can obtain an AES that contains different adversarial examples with diverse deep features to achieve the augmentation of adversarial examples. Given the diversity of the generated adversarial examples in the AES of the surrogate model, the optimal adversarial example for a certain target model is likely contained in our generated AES. Thus, the generated AES is expected to have high-transferability. In order to find the optimal adversarial example of a specific target model in the AES, we use the query method to achieve this goal. Experimental results showcase the superiority of the proposed AEA framework for black-box attack in two representative BIS tasks including salient object detection and camouflage object detection.

1. Introduction

The objective of binary image segmentation (BIS) is to segment the objects of interest from the background in an image, highlighting the foreground target regions. During the past decade, deep neural network (DNN) has demonstrated remarkable performance in this task [1,2]. However, it has been demonstrated that DNN-based BIS models are susceptible to adversarial attacks [3]. Conceptually, adversarial attacks involve adding malicious perturbations to benign inputs to generate adversarial examples. These adversarial examples can mislead the BIS models, rendering the targets in the images undetectable by the model. Therefore, research on adversarial attacks can verify the security of DNN-based BIS models, help promote the development of BIS models towards higher robustness and reliability.

Adversarial attacks are mainly divided into white-box and black-box attacks [4,5]. White-box attacks represent that the attacker can obtain the internal information of the target model, and the attacker can use gradient information to generate adversarial examples. In the black-box setting, the attacker has no or limited knowledge about the model. In practical situation, the attacker often does not have full access to the target model, so the research on black-box attack is more meaningful.

In the context of black-box attacks, estimating the gradient of the target model is a common idea, and broadly there are two categories: i.e., query-based attacks [6,7] and transfer-based attacks [8]. The former generates adversarial examples by querying the output of the target model and estimating the gradient of the target model using optimization methods such as random gradient estimation and finite difference. The latter generates adversarial examples for surrogate models to achieve the goal of attacking target model. However, query-based attacks usually require a large number of queries, while transfer-based attacks do not require multiple accesses to the model, thus completely avoiding suspicion and being more suitable for practical applications.

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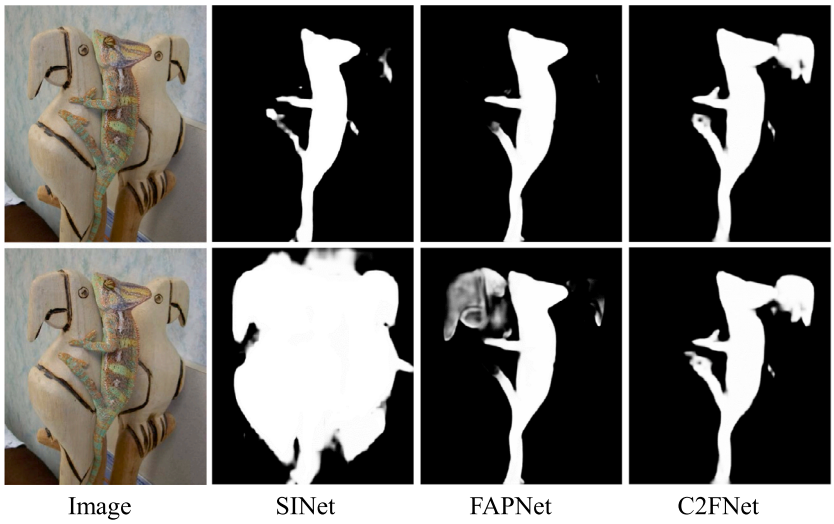

Fig. 1. The prediction results of SINet [11], FAPNet [12], and C2FNet [13] with the original image and the adversarial example as input, respectively. The first row shows the original image and its corresponding prediction results. The second row shows the adversarial example generated for SINet and its corresponding prediction results.

Although many transfer-based attacks have been proposed [9,10], their transferability remains unsatisfactory due to the differences between the surrogate model and the target model and the differences in gradient information within the model can lead to poor transferability, especially when ensuring that adversarial examples maintain good visual quality. As shown in Fig. 1, adversarial examples generated for SINet [11] (surrogate model) do not work well on FAPNet [12] and C2FNet [13] (target models). In addition, existing black-box attack methods is limited to generate only one adversarial example, which cannot well adapted to diverse target models.

To alleviate this limitation, we propose a novel adversarial example augmentation (AEA) framework to improve the transferability of black-box attacks. Different from the existing black-box attack methods which are limited to generate a single adversarial example for the target model or the surrogate model, our method generates an adversarial example set (AES) containing a set of distinct adversarial examples. Specifically, we first employ an existing model as the surrogate model which is attacked to optimize the adversarial perturbation via maximizing the BCE loss between the prediction of the surrogate model and the pseudo label, thus producing a sequence of adversarial examples. During the optimization process, besides the Binary Cross-Entropy (BCE) loss, we additionally introduce deep feature losses among different adversarial examples to fully distinguish the generated adversarial examples. In this way, we can obtain an AES that contains different adversarial examples with diverse deep features to achieve the augmentation of adversarial examples. Given the diversity of the generated adversarial examples in the AES of the surrogate model, the optimal adversarial example for a certain target model is likely contained in our generated AES. Thus, the generated AES is expected to have high-transferability. In order to find the optimal adversarial example of a specific target model in the AES, we use the query method to achieve this goal. The required number of queries is just the number of adversarial samples contained in AES. Overall, the main contributions of this paper are as follows:

1. We propose a novel AEA framework which focuses on generating an AES rather than a single adversarial example for a surrogate model towards high-transferability black-box attack. To the best of our knowledge, this is the first work that generates multiple adversarial examples with distinctive deep features to effectively improve the transferability.
2. We propose an AES generator that generates a set of distinctive adversarial examples by optimizing: (i) the BCE loss between the prediction of the surrogate model and the pseudo label, (ii) the deep feature losses among different adversarial examples. In addition, we adopt a query method to attack the target model by finding the optimal adversarial example from the AES.
3. We introduce the HDR-VDP [14] metric to measure the perturbation visibility of the generated adversarial examples for the first time. Under the condition of similar perceptual quality score (i.e., the HDR-VDP score), our proposed AEA method has shown superior attack performance over the state-of-the-art methods in a wide range of experiments, showcasing its high-transferability.

2. Related works

2.1. Adversarial attack

With the continuous development of deep learning, DNN has shown amazing results in the field of images [15–17], and even videos [18,19]. But its nature of being easily disturbed by small perturbations has not been addressed. In order to explore the robustness and security of deep learning models under different tasks, adversarial attack methods for different fields have also been generated.

2.2. White-box attack

White-box attacks [20,21] is an attack method that produces an adversarial example with knowledge of the internal information of the target model. The simplest and most direct attack method is to use a first-order optimizer (random gradient descent, Adam *etc.*) or a second-order optimizer (Newton method or quasi-Newton method *etc.*) to reverse gradient propagation of the final output design objective function of the target model to optimize against perturbations.

The earliest adversarial perturbations were generated through white-box attacks based on quasi-Newton methods, as proposed by Szegedy et al. in 2014. Subsequently, Goodfellow et al. [22] proposed the Fast Gradient Sign Method (FGSM), a one-step gradient update method for calculating norm-bounded perturbations. This method emphasizes the efficiency of perturbation calculation rather than achieving high fooling rates. FGSM played a significant role in driving subsequent research on adversarial attacks, leading to various improved methods based on FGSM. Kurakin et al. [23] analyzed l-norm normalization and extended FGSM to the Iterative Fast Gradient Sign Method (I-FGSM), a rapidly iterative gradient-based method that gradually adds perturbations with smaller step sizes. In each iteration, perturbations are added, and gradients are computed using the adversarial examples generated at the

current step, updating the adversarial examples with gradient information. Dong et al. [24] proposed Momentum Iterative Fast Gradient Sign Method (MI-FGSM), adding momentum to the iterative FGSM method to further enhance the transferability of perturbations.

In addition to FGSM-based variant attack methods, there are many other attack methods. Su et al. [25] proposed the one pixel attack, which employs differential evolution algorithms to gradually eliminate pixels with low relevance, ultimately identifying a crucial pixel. Modifying this pixel can disrupt the classifier's decision. Dong et al. [26] proposed an attention mechanism to generate perturbations guided by superpixels around objects in the image, achieving the attack objective. Omid et al. [27] proposed a generative adversarial network framework to train a generator to generate universal perturbations against the target network. This method has shown effectiveness in the fields of classification, recognition, and segmentation. White-box attacks require access to the complete information of the model, a condition often unattainable in practical applications. This constraint limits the applicability of white-box attacks in the actual situation where obtaining full information about the target model is not feasible.

2.3. Black-box attack

The black-box attack [28] method is designed to target a model without having access to its internal information, so the black-box attack method aligns more closely with practical situations.Black-box attack methods are predominantly categorized into query-based attacks and transfer-based attacks.

2.3.1. Query-based attack

In query-based attacks, these methods relies solely on the output results of the target model, iteratively obtaining the final adversarial example through repeated queries and optimization. Minimizing the number of queries while achieving optimal attack performance presents a challenge for query-based attack methods. Rahmati et al. [29] proposed a framework based on the geometry of decision boundaries, launching black-box attacks with minimal queries by querying the target model a small number of times and only obtaining the top-1 label. Li et al. [7] proposed the Query Efficient Boundary-based Black-box Attack (QEBA) method, iteratively adding perturbations to the source image to retain its original label while altering the image to form a perceptually clean target image of a different object. Cheng et al. [30] also presented a query-efficient attack, modifying the optimization objective from their previous work to use query results for binary search to estimate the gradient of the target model [31]. They significantly reduced the number of queries by focusing on estimating the gradient sign rather than the gradient, thus improving the attack.

2.3.2. Transfer-based attack

The transfer-based attacks primarily engages with the surrogate model, where the internal information of the surrogate model is entirely accessible. It executes a white-box attack on the surrogate model, ensuring that the generated adversarial example can effectively target the intended model. The primary objective of this attack method is to accentuate the inherent transferability of the perturbation, leading to the emergence of various strategies designed for this purpose. Lin et al. utilized Nesterov gradient acceleration in an I-FGSM to compute more generalized and transferable perturbations. The authors also introduced a scale-invariant attack method, which induces a model ensemble from the original model using data transformations that retain the original model's loss. Adversarial instances computed using these models exhibited enhanced transferability. Wu et al. [9] proposed an attack that enhances the transferability of perturbations by concentrating the perturbations more on salient image regions. Huang et al. [10] fine-tuned adversarial examples using representations from a pre-specified layer of the source model to improve attack transferability. Li et al. [32] addressed two characteristics limiting the transferability of white-box target attacks, making gradient-based target attacks more transferable. The first characteristic involves the reduction in gradient magnitude during the iteration process, leading to noise elimination, and was addressed in [32] by allowing adaptive gradient sizes during optimization. The second characteristic, closeness between adversarial examples and real-class regions, was mitigated through regularization based on metric learning.

2.4. Attack methods for BIS

The representative BIS task includes salient object detection (SOD), camouflaged object detection (COD), co-salient object detection (CoSOD), and more. The common objective is to segment target objects in a scene from the background, emphasizing the target area through binarization. Currently, there are limited attack and defense methods for BIS tasks.

Che et al. [3] proposed a sparser adversarial perturbation in the feature space specifically targeting deep visual saliency models. This method requires a certain amount of model information. Subsequently, they incorporated query attacks into their original method to extend the attack to a broader range of SOD models. Gao et al. [33] proposed a perturbation generation framework for CoSOD, simultaneously generating exposure perturbations and noise perturbations that affect the output of the CoSOD model. Piao et al. [34] addressed weakly supervised SOD by designing a noise-robust adversarial learning framework. They proposed a noise-sensitive training strategy within this framework, enhancing the robustness of weakly supervised SOD networks to noise information in pseudo-labels.

Many existing attack methods for BIS tasks primarily target the backbone network of the model. They utilize the backbone network as a surrogate model and conduct attacks on the target model through specific operations. However, the backbone network structure of current models for BIS tasks varies. Consequently, merely relying on a backbone network as the surrogate model is insufficient. To address this limitation, we propose an adversarial example enhancement framework, which uses the existing model as a surrogate model to maximize the gap between prediction results and pseudo labels, maintain the uniqueness of adversarial examples, generate AES, and conduct query attacks on the target model. In addition to this, the image quality of the adversarial examples is also important. We adopted relevant metrics in the field of image quality assessment to evaluate the adversarial examples [35–37]. Experimental results demonstrate the superior attack efficacy of our proposed method compared to the comparison method.

3. Proposed method

3.1. Overview

As illustrated in Fig. 2, our proposed AEA framework involves an AES generator and a query-based attacker. Given an original image I, instead of generating only a single adversarial example for the surrogate model, our main idea is to generate a set of distinctive adversarial examples (AES) $\mathbb{A} = \{A_i | i = 1, 2, \dots, N\}$ for the surrogate model to best increase the probability of containing the optimal adversarial example for an arbitrary target model, thus improving the attack transferability. Specifically, the first adversarial example A_1 is generated by optimizing the BCE loss between the prediction result of the surrogate model and the pseudo label. The subsequent adversarial examples A_i are generated by optimizing the same BCE loss as well as a deep feature loss which aims at keeping the feature-level distinctiveness between A_i and all previous adversarial examples $\{A_1, A_2, \dots, A_{i-1}\}$. Specifically, A_i is compared with $\{A_1, A_2, \dots, A_{i-1}\}$ using the Kullback–Leibler (KL) divergence metric in terms of the hierarchical deep features which are extracted by multiple pre-trained deep backbone networks. With the generated AES $\mathbb{A} = \{A_i | i = 1, 2, \dots, N\}$, we finally apply a query-based method to seek for the optimal adversarial example for an arbitrary target model. The number of queries is just the number of adversarial samples in AES.

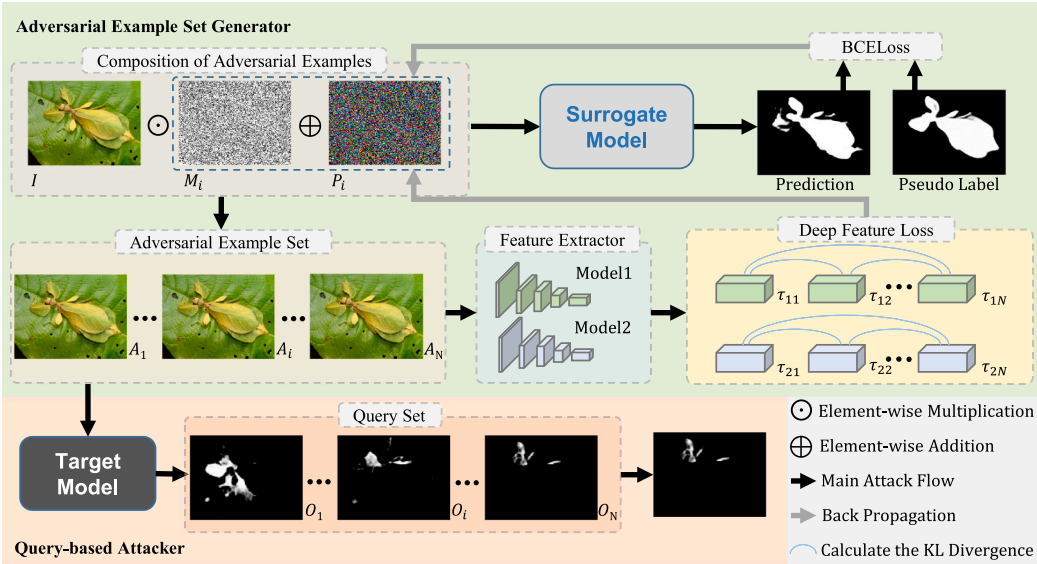

Fig. 2. Overview of the AEA framework. Our attack methodology consists of two key components: AES generator and query-based attacker. First, we utilize an AES generator to create a series of unique adversarial examples that enable the surrogate model to make false prediction results. Each adversarial example is obtained by applying multiplicative and additive perturbations to the original image. The perturbation is updated by enlarging the difference between the prediction results of the proxy model and the pseudo label of the images, and the uniqueness of the adversarial examples is guaranteed by limiting the difference between them. Finally, the query-based attacker queries the target model using AES to obtain the optimal adversarial example.

3.2. Adversarial example set generator

In the AES generator, the original image is denoted as I, and the AES is represented as $\mathbb{A} = \{A_i | i = 1, 2, \ldots, N\}$, where N is the number of generated adversarial examples. The adversarial example A_i is an altered version of I:

$$A_i = C_{RGB}((C_{YUV}(I)) \cdot M_i) + P_i, \tag{1}$$

where C_{RGB} and C_{YUV} respectively represent the conversion of image to the RGB color space and the YUV color space, M_i and P_i respectively represent the multiplicative and additive perturbations to the adversarial example A_i. The multiplicative perturbation M_i specifically operates on the Y channel of I in the YUV color space, inducing pixel-level luminance disturbances without altering the overall color of the image. In contrast to [38], the pixel-level luminance mask perturbation employed here avoids causing global luminance fluctuations in adversarial example and prevents the formation of localized dark patches, thereby preserving the overall quality of the adversarial example. The objective of the AES generator is to generate a set of adversarial examples which can cause the surrogate model to make incorrect predictions while the added perturbations are perceptually invisible. Therefore, the optimization objective of our attack method can be formulated as follows:

$$\mathcal{L}_{img} = -\mathcal{L}_{BCE}(\mathcal{S}(A_i), PL) + \lambda_1 \mathcal{L}_{norm}(A_i, I), \tag{2}$$

where $\mathcal{L}_{BCE}$ is the classical BCE loss, $\mathcal{L}_{norm}$ is implemented by ℓ_∞-norm induced metrics, $\mathcal{S}$ denotes the surrogate model, PL stands for pseudo label which is obtained by the current most advanced model, λ_1 is the parameter used to control the relative importance of the two terms. By minimizing the above loss, the prediction result of the surrogate model is encouraged to be different with the pseudo label and the perceptual quality of the generated adversarial example is enforced to be similar to the quality of the original image.

The adversarial examples generated in the above process may fall into local optima of the surrogate model, resulting in the generated adversarial examples becoming increasingly similar. To ensure the distinctiveness of the generated adversarial examples in the AES, it is necessary to introduce additional constraints to achieve this goal. Given the powerful capability of deep neural networks of extracting hierarchical feature representations of images, it is important to distinguish each generated adversarial example in terms of their corresponding hierarchical deep features. With this insight, besides the above loss $\mathcal{L}_{img}$, we also propose to optimize a depth feature loss to keep the feature-level distinctiveness between A_i and all previous adversarial examples $\{A_1, A_2, \ldots, A_{i-1}\}$ during the optimization process. The deep feature loss is formulated as follows:

$$\mathcal{L}_{df}(A_i, A_j) = \left| \min(\mathcal{KL}(\mathcal{F}(A_i), \mathcal{F}(A_j))) - \gamma \right|, \quad j = 1, 2, \ldots, i-1, \tag{3}$$

where A_i stands for the currently generated adversarial example, A_j stands for the previous adversarial examples, $\mathcal{F}$ represents the deep feature extractor, $\mathcal{KL}$ denotes a function that calculates the KL divergence [39], γ is the pre-defined distance value. In the implementation, we select half of the feature layers at equal intervals to calculate the KL divergence, as illustrated in Fig. 3. In addition, we adopt ResNet and Res2Net [40] with the fully connected layers to be removed as the feature extractor $\mathcal{F}$. The deep feature loss will not be used when generating the first adversarial example. Starting from the generation of the second adversarial example, a deep feature loss needs to be used to guarantee the feature difference between different adversarial examples. By minimizing the above deep feature loss, the minimum distance between the previous adversarial examples and the currently generated adversarial example is encouraged to be close to γ. There will be a discrepancy between the current generated adversarial example and each previous adversarial example.

To facilitate understanding, we further illustrate the whole process of AES generation in Fig. 4. In the figure, $\mathcal{S}$ represents all possible adversarial examples that are effective to the surrogate model and T_i represents all possible adversarial examples that are effective to a certain target model. The intersection region of $\mathcal{S}$ and T_i represents the adversarial examples that are both effective to the surrogate model and the target model. Our goal is to start with an initial adversarial example A_1 and progressively obtain more diverse adversarial examples, such as A_2 and A_3, by simultaneously optimizing $\mathcal{L}_{img}$ and $\mathcal{L}_{df}$, thereby gradually expanding our AES.

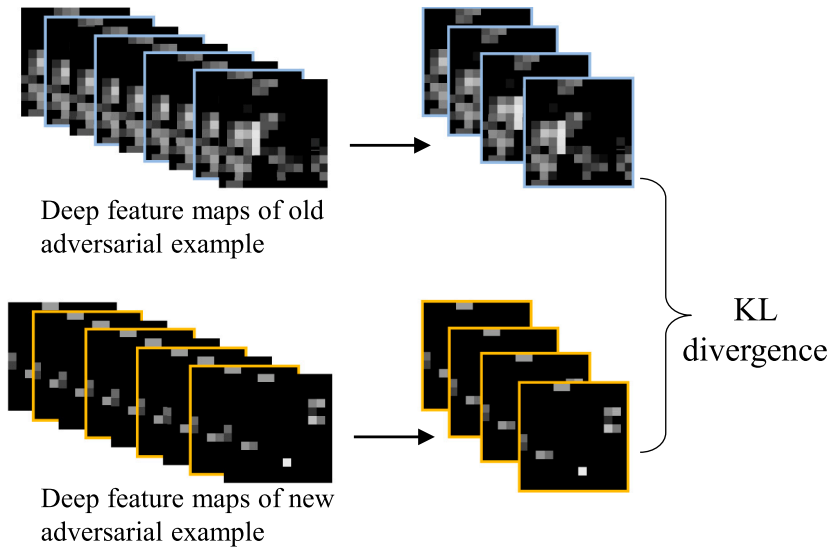

Fig. 3. Illustration of KL divergence computation between old and new adversarial examples. The deep feature maps of adversarial examples are encoded by the feature extractor. When calculating the KL divergence, an equally spaced extraction of half of the feature layers is performed for KL divergence computation.

Algorithm 1 Adversarial Example Augmentation.

Input:
Original image I; Multiplicative perturbation M;
Additive perturbation P; Pseudo label PL;
Surrogate model $\mathcal{S}$; Deep feature extractor $\mathcal{F}$;
Index of the current example i;
Index of previously generated examples j;
The number of augmentation α;
Maximum number of iterations β;
Perturbation feature divergence constraint γ;
Regulation parameters λ_1, λ_2;
Similarity measure $\mathcal{D}$; Limits on similarity φ;
The allowed bias δ;
Output:
Adversarial example set $\mathbb{A} = \{A_1, A_2, \dots, A_N\}$

```
while (m < α) do
    M ← 1, P ← 0;
    while (n < β) do
        A_i ← I × M_i + P_i;
        L_img ← L_BCE(S(A_i), PL) + λ_1 ||(A_i, I)||_∞;
        if (𝔸 = ∅) then
            Update A_i by optimizing L_img;
        else L_df ← | min KL(F(A_i), F(A_j)) − γ |;
            Update A_i by optimizing L_img + λ_2 L_df;
        end if.
        if (D(S(A_i), PL) < φ and L_df < δ then
            𝔸 append A_i;
        else n ← n + 1;
        end if.
    end while.
    m ← m + 1;
end while.
return 𝔸 = {A_1, A_2, ..., A_N}
```

The detailed processes are described as follows. First, we generate the initial adversarial example A_1 against the surrogate model by optimizing $\mathcal{L}_{img}$, as defined in Eq. (2). Subsequent adversarial examples are generated by optimizing $\mathcal{L}_{img}$ and $\mathcal{L}_{df}$ simultaneously, i.e., $\mathcal{L}_{img}(I, PL, A_i) + \lambda_2 \mathcal{L}_{df}(A_i, A_j),\ i > 1$. Take the generation of A_i as an example, we calculate the KL divergence distance $l_{i,j}$ between the current adversarial example A_i and each old adversarial example A_j and then find the old adversarial example with the minimum distance $\hat{l}_i = \min\{l_{i,j}, j = 1, 2, \dots, i-1\}$ which is encouraged to be a fixed value γ. However, the examples strictly locate in the circle with radius γ may be difficult to obtain. Therefore, we relax the distance γ by a small bias $\pm\delta$, forming an annular region. Once the generated adversarial example falls within this annular region, it will be recorded as a new adversarial example, and the next round of optimization begins with this current adversarial example as the start point. In this way, different adversarial examples can be generated and the augmentation is completed when the desired number of adversarial examples are generated. To facilitate understanding, we illustrate the algorithm details in Algorithm 1 where $\mathcal{D}$ stands for the operation of calculating the Intersection over Union (IoU).

3.3. Query-based attacker

After obtaining the AES $\mathbb{A}$ through the above steps, we carry out query attacks on the target model. We utilize the query method to obtain the best performing adversarial examples in the AES. By comparing the results after querying, we can obtain the optimal adversarial example. The number of queries is just the number of adversarial examples in the AES. The attack results can be summarized as follows:

$$A^* = \arg\max_{A_i} \mathcal{L}_{BCE}(\mathcal{T}(A_i), PL), \tag{4}$$

where $\mathcal{T}$ represents the target model. The adversarial examples A_i are sequentially used as target model inputs to obtain the final prediction result set $\{\mathcal{T}(A_i)\}$. Each prediction result $\mathcal{T}(A_i)$ is compared to the pseudo label PL to find the optimal adversarial example A^*. This process is independent of the adversarial example generation process described above and does not participate in the process of adversarial example generation.

4. Experiments

4.1. Datasets and comparison models

4.1.1. Datasets

We conduct experiments on three datasets, i.e., ECSSD [41], CAMO [42], and COD10K [11]. ECSSD consists of 1000 images which are all used for attack testing on the SOD task. CAMO and COD10K contain 250 and 2000 testing images, respectively, which are employed for attack testing on the COD task.

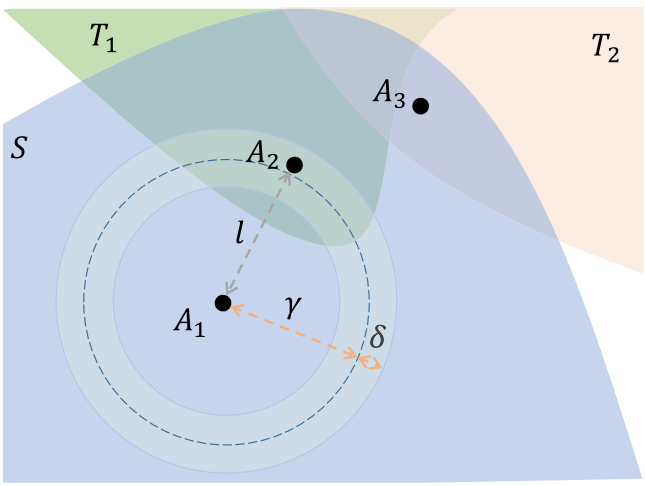

Fig. 4. Illustration of the process of AES generation. S represents all possible adversarial examples that are effective to the surrogate model and T_1, T_2 represents all possible adversarial examples that are effective to the different target models. The intersection region of S and T_1, T_2 represents the adversarial examples that are both effective to the surrogate model and the target model.

4.1.2. Models

For the SOD task, F3Net [43] is chosen as the surrogate model, while PSGLoss [44], BASNet [45], EGNet [46], and BCMNet [47] are selected as target models, and the pseudo label is the prediction result of PFSNet [48] on the original image. For the COD task, SINet [11] is chosen as the surrogate model, while C2FNet [13], PFNet [49], FAPNet [12], and SegMaR [50] are chosen as target models, and the pseudo label is the prediction result of FSPNet [51] on the original image.

4.1.3. Comparison methods

The comparison methods include FGSM [22], I-FGSM [23], and MI-FGSM [24]. Additionally, we use ℓ_1-norm and BCE losses as optimization objectives. We use the comparison methods to carry out white-box attacks against F3Net and SINet, and use the obtained adversarial examples to launch attacks against other target models.

4.2. Evaluation metrics

We employ four commonly-used evaluation metrics for BIS tasks to assess the effectiveness of attacks. These metrics include Mean Absolute Error (M) [52], Structural Similarity (S_α) [53], Weighted F-measure (F_β^ω) [54], and Average Precision (AP) [52]. Specifically, M is used to measure pixel-level similarity between the prediction result and the ground truth. S_α measures the structural similarity between the prediction result and the ground truth. F_β^ω combines recall and precision, eliminating the equal consideration of each pixel's impact found in traditional metrics. AP measures the accuracy between the prediction result and the ground truth. When comparing attack effectiveness, the goodness of these metrics is opposite to that in BIS tasks, i.e., poorer metrics indicate better attack performance.

4.3. Implementation details

The framework is implemented in PyTorch, utilizes the Adam optimizer to optimize multiplicative perturbation and additive perturbation. The learning rate for multiplicative perturbation is set to 1e−33, while the learning rate for additive perturbation is set to 2e−3. The limits on similarity φ is set to 0.2. The maximum number of adversarial samples α in AES and the maximum number of iterations β is set to 20 and 200. The perturbation feature divergence constraint γ and bias δ is set to 500 and 250. The parameter λ_1 in Eq. (2) is set to 4. For the setting of γ and δ, it should be noted that when γ constant δ is too small, the adversarial perturbation is subject to large constraints, which will lead to too few pictures in the AES and reduce the diversity of adversarial examples. When δ is similar to γ, the adversarial perturbation is subject to small constraints. Although it can quickly generate more pictures, the feature gap between adversarial examples is small, which will also reduce the diversity of adversarial examples. And the size of γ then directly affects the number of adversarial examples generated.

4.4. Comparison results

4.4.1. Quantitative comparison

We present the metrics of the AEA and those of other comparison methods in Table 1. It can be seen that our method achieves the optimal results in the tests of most target models, and also achieves good scores for PSGLoss in SOD task and PFNet in COD task. This indicates the versatility of our AEA approach across both SOD and COD tasks. Through the analysis of scores, surrogate models, and target models, the following conclusions can be drawn: on target models with significant differences networks (e.g. comparing F3Net with BCMNet, SINet with FAPNet, where F3Net and SINet share a ResNet [55] backbone, BCMNet involves both ResNet and transformer [56], and FAPNet employs Res2Net [40] as the backbone), our method consistently outperforms the comparison methods, signifying superior transferability. To visually highlight the advantages of our proposed AEA method, Fig. 5 presents the F-measure curves for BIS models under different attack methods. From Fig. 5, it is evident that our proposed method leads to the most significant reduction in the model's predictive performance.

4.4.2. Perturbation visibility comparison

In terms of the visual effect of the perturbations, we use HDR-VDP [14] to quantify the perturbation visibility of adversarial examples generated by our AEA and other comparison methods. The average HDR-VDP scores of adversarial examples generated by different methods are shown in Table 2. It can be seen that the average HDR-VDP scores of adversarial examples generated by our AEA method are even slightly higher than those of the comparison methods, indicating the perturbations induced by our method are less visible to human perception. Fig. 6 shows the adversarial examples generated by different attack methods. As can be seen from Fig. 6, the visual effects of these adversarial examples are very similar, without any obvious artifacts.

4.4.3. Qualitative comparison

In order to further demonstrate the effectiveness of our AEA method, we also conduct qualitative comparisons. Similarly, under the HDR-VDP scores in Table 2, we present the attack results of various methods on BIS tasks. In Fig. 7, the first five rows show the prediction results of BASNet for adversarial examples generated by different attack methods on the ECSSD [41] dataset, while the last five rows show the prediction results of FAPNet for adversarial examples generated by different attack methods on the CAMO [42] and COD10K [11] datasets. It can be observed from the results that among various attack methods, our proposed AEA has the greatest impact on the prediction results. This method significantly deviates the prediction results from the GT images and the original predicted results. Our method can achieve better attack performance even with slightly higher HDR-VDP quality scores than other attack methods, thus demonstrating superior effectiveness. According to the results shown in Table 1, Figs. 5 and 7, it can be concluded that our proposed attack method has better transferability compared with other comparison methods.

Table 1
In BIS tasks, comparisons were made among different models using various comparison methods on the ECSSD dataset, as well as the test sets of CAMO and COD10K. the best results are highlighted in **Bold**. We evaluate the metric with ap, m, s_α, f_β^ω. ↑ means that a higher score corresponds to better attack effectiveness, ↓ means a lower score corresponds to better attack effectiveness.

ECSSD [41]																
Comparison method	PSGLoss [44]				BASNet [45]				EGNet [46]				BCMNet [47]			
	AP ↓	M ↑	S_α ↓	F_β^ω ↓	AP ↓	M ↑	S_α ↓	F_β^ω ↓	AP ↓	M ↑	S_α ↓	F_β^ω ↓	AP ↓	M ↑	S_α ↓	F_β^ω ↓
Original	0.926	0.113	0.719	0.688	0.913	0.056	0.870	0.839	0.936	0.037	0.925	0.904	0.926	0.099	0.782	0.737
Gaussian noise	0.924	0.118	0.706	0.671	0.906	0.064	0.854	0.816	0.935	0.040	0.919	0.894	0.909	0.088	0.806	0.775
$FGSM_{BCE}$ [22]	0.847	0.126	0.698	0.649	0.898	0.061	0.862	0.829	0.890	0.046	0.906	0.872	0.879	0.106	0.769	0.722
$FGSM_{L1}$ [22]	0.864	0.119	0.712	0.669	0.902	0.057	0.870	0.839	0.894	0.042	0.913	0.881	0.888	0.099	0.783	0.741
I-$FGSM_{BCE}$ [23]	**0.406**	**0.338**	**0.450**	**0.334**	0.894	0.073	0.831	0.785	0.908	0.060	0.882	0.838	0.864	0.122	0.728	0.662
I-$FGSM_{L1}$ [23]	0.470	0.294	0.505	0.390	0.897	0.071	0.837	0.793	0.907	0.055	0.890	0.848	0.875	0.116	0.739	0.680
MI-$FGSM_{BCE}$ [24]	0.954	0.117	0.696	0.665	0.933	0.051	0.882	0.855	0.961	0.032	0.936	0.919	0.942	0.085	0.815	0.788
MI-$FGSM_{L1}$ [24]	0.803	0.177	0.558	0.442	0.913	0.068	0.845	0.806	0.936	0.059	0.885	0.849	0.873	0.120	0.734	0.671
Ours	0.436	0.300	0.481	0.354	**0.874**	**0.086**	**0.802**	**0.742**	**0.876**	**0.073**	**0.855**	**0.795**	**0.799**	**0.146**	**0.668**	**0.573**
CAMO [42]																
Comparison method	PFNet [49]				FAPNet [12]				C2FNet [13]				SegMaR [50]			
	AP ↓	M ↑	S_α ↓	F_β^ω ↓	AP ↓	M ↑	S_α ↓	F_β^ω ↓	AP ↓	M ↑	S_α ↓	F_β^ω ↓	AP ↓	M ↑	S_α ↓	F_β^ω ↓
Original	0.802	0.087	0.774	0.687	0.805	0.078	0.811	0.727	0.794	0.084	0.787	0.702	0.781	0.115	0.729	0.594
Gaussian noise	0.798	0.096	0.737	0.627	0.797	0.082	0.803	0.714	0.788	0.089	0.777	0.684	0.775	0.116	0.724	0.585
$FGSM_{BCE}$ [22]	0.566	0.161	0.593	0.409	0.675	0.125	0.699	0.554	0.667	0.131	0.659	0.506	0.578	0.165	0.592	0.388
$FGSM_{L1}$ [22]	0.569	0.164	0.583	0.394	0.678	0.126	0.695	0.550	0.665	0.134	0.651	0.495	0.576	0.166	0.583	0.376
I-$FGSM_{BCE}$ [23]	0.263	**0.306**	0.395	0.161	0.682	0.126	0.719	0.572	0.668	0.131	0.654	0.492	0.373	0.222	0.494	0.230
I-$FGSM_{L1}$ [23]	**0.251**	0.296	**0.387**	**0.146**	0.688	0.123	0.719	0.573	0.678	0.130	0.654	0.488	0.355	0.222	**0.484**	**0.217**
MI-$FGSM_{BCE}$ [24]	0.918	0.033	0.906	0.874	0.865	0.052	0.873	0.813	0.851	0.060	0.846	0.782	0.893	0.088	0.814	0.721
MI-$FGSM_{L1}$ [24]	0.921	0.031	0.910	0.879	0.862	0.051	0.875	0.815	0.852	0.058	0.845	0.782	0.897	0.088	0.813	0.721
Ours	0.543	0.181	0.589	0.403	**0.619**	**0.151**	**0.674**	**0.510**	**0.582**	**0.164**	**0.603**	**0.410**	**0.351**	**0.236**	0.485	0.224
COD10K [11]																
Comparison method	PFNet [49]				FAPNet [12]				C2FNet [13]				SegMaR [50]			
	AP ↓	M ↑	S_α ↓	F_β^ω ↓	AP ↓	M ↑	S_α ↓	F_β^ω ↓	AP ↓	M ↑	S_α ↓	F_β^ω ↓	AP ↓	M ↑	S_α ↓	F_β^ω ↓
Original	0.693	0.043	0.788	0.637	0.714	0.041	0.808	0.666	0.695	0.043	0.789	0.642	0.681	0.061	0.751	0.559
Gaussian noise	0.689	0.046	0.767	0.605	0.708	0.042	0.800	0.653	0.685	0.046	0.779	0.625	0.676	0.061	0.750	0.556
$FGSM_{BCE}$ [22]	0.496	0.090	0.680	0.454	0.625	0.058	0.759	0.578	0.599	0.063	0.733	0.545	0.506	0.089	0.674	0.415
$FGSM_{L1}$ [22]	0.527	0.081	0.702	0.490	0.644	0.053	0.771	0.600	0.617	0.059	0.746	0.566	0.536	0.084	0.689	0.443
I-$FGSM_{BCE}$ [23]	0.374	0.143	0.576	0.286	0.629	0.062	0.748	0.554	0.623	0.060	0.715	0.517	0.439	0.114	0.614	0.299
I-$FGSM_{L1}$ [23]	0.386	0.139	0.594	0.308	0.624	0.062	0.751	0.556	0.614	0.060	0.715	0.514	0.442	0.114	0.622	0.307
MI-$FGSM_{BCE}$ [24]	0.706	0.048	0.756	0.590	0.733	0.041	0.804	0.661	0.717	0.043	0.783	0.634	0.683	0.064	0.736	0.528
MI-$FGSM_{L1}$ [24]	0.527	0.088	0.646	0.398	0.672	0.054	0.768	0.596	0.665	0.055	0.741	0.565	0.530	0.090	0.658	0.381
Ours	**0.152**	**0.287**	**0.419**	**0.127**	**0.518**	**0.093**	**0.684**	**0.444**	**0.471**	**0.100**	**0.640**	**0.387**	**0.236**	**0.184**	**0.502**	**0.166**

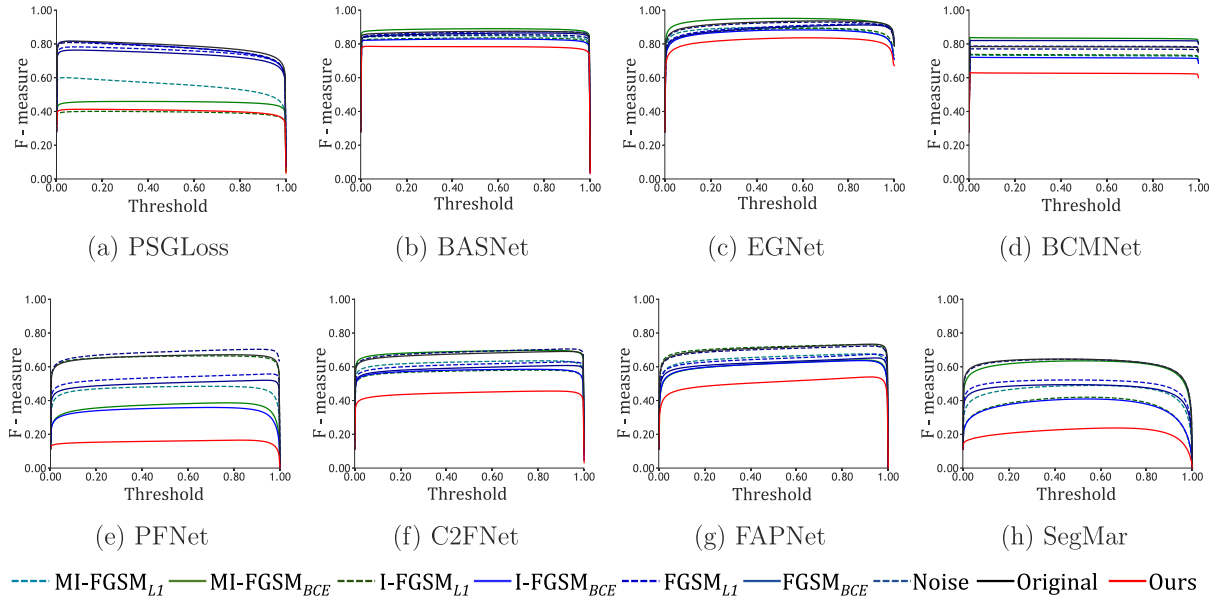

Fig. 5. F-measure curves of different network models on different AESs. The first row shows curves for various network models under the SOD task on the ECSSD dataset, while the second row displays curves for different network models under the COD task on the test sets of CAMO and COD10K.

Table 2
The average HDR-VDP scores of adversarial examples generated by different attack methods. In addition, SSIM, PSNR is also used for a more rigorous contrast of images.

Comparison method	CAMO [42]			COD10K [11]			ECSSD [41]		
	HDR-VDP	SSIM	PSNR	HDR-VDP	SSIM	PSNR	HDR-VDP	SSIM	PSNR
Gaussian noise	8.349	0.930	26.221	8.035	0.8943	26.324	7.916	0.894	26.138
$FGSM_{BCE}$ [22]	8.153	0.925	32.897	8.234	0.933	35.185	7.907	0.917	33.373
$FGSM_{L1}$ [22]	8.153	0.925	32.897	8.233	0.933	35.185	7.907	0.917	33.373
$I\text{-}FGSM_{BCE}$ [23]	8.349	0.938	33.479	8.362	0.942	35.290	7.917	0.917	32.720
$I\text{-}FGSM_{L1}$ [23]	8.349	0.938	33.495	8.293	0.938	34.891	7.920	0.918	32.726
$MI\text{-}FGSM_{BCE}$ [24]	8.351	0.937	33.661	8.413	0.945	35.610	7.951	0.919	32.809
$MI\text{-}FGSM_{L1}$ [24]	8.353	0.937	33.661	8.411	0.945	35.630	7.943	0.918	33.002
Ours	8.364	0.937	33.948	8.414	0.949	36.505	7.973	0.919	33.813

Table 3
In the Co-SOD task, comparisons were made among different models using various comparison methods on the CoCA [60] dataset. The best results are highlighted in **Bold**. We evaluate the metric with AP, M, S_α, F_β^ω. ↑ means that a higher score corresponds to better attack effectiveness, ↓ Means A Lower score corresponds to better attack effectiveness.

Comparison method	GICD [57]				CoRP [58]				GCoNet [59]			
	AP ↓	M ↑	S_α ↓	F_β^ω ↓	AP ↓	M ↑	S_α ↓	F_β^ω ↓	AP ↓	M ↑	S_α ↓	F_β^ω ↓
Original	0.557	0.116	0.679	0.525	0.555	0.107	0.711	0.537	0.655	0.083	0.724	0.542
Noise	0.556	0.12	0.667	0.505	0.382	0.142	0.576	0.334	0.617	0.116	0.595	0.349
$FGSM_{BCE}$ [22]	0.520	0.13	0.664	0.493	0.394	0.152	0.576	0.339	0.589	0.117	0.592	0.34
$FGSM_{L1}$ [22]	0.528	0.125	0.673	0.505	0.401	0.154	0.581	0.354	0.59	0.117	0.592	0.34
$I\text{-}FGSM_{BCE}$ [23]	0.51	0.134	0.671	0.471	**0.358**	0.144	0.562	0.297	0.614	0.117	0.594	0.335
$I\text{-}FGSM_{L1}$ [23]	0.524	0.13	0.662	0.492	0.434	0.133	0.588	0.347	0.605	0.117	0.602	0.347
$MI\text{-}FGSM_{BCE}$ [24]	0.577	0.117	0.679	0.531	0.421	0.134	0.601	0.374	0.631	0.115	0.599	0.353
$MI\text{-}FGSM_{L1}$ [24]	0.522	0.133	0.675	0.474	0.455	0.131	0.574	0.315	0.615	**0.119**	**0.581**	**0.323**
Ours	**0.502**	**0.136**	**0.649**	**0.47**	0.375	**0.163**	**0.558**	**0.293**	**0.588**	**0.119**	0.591	0.353

Fig. 6. Perturbation visibility comparison of adversarial examples generated by different attack methods. The first and second columns are examples selected from the SOD dataset, while the third and fourth columns are examples selected from the COD dataset.

4.4.4. Generalization capability

To demonstrate the generalization capability of the proposed attack method, we also conduct experiments on the Co-SOD task. In this experiment, F3Net was used as the surrogate model, while GICD [57], CoRP [58], and GCoNet [59] are served as the target models. The CoCA [60] dataset was utilized for evaluation. The quantitative comparison results, presented in Table 3, show that our proposed method also achieves relatively good performance on the Co-SOD task. This also proves that our method has a certain generalization capability.

4.5. Ablation studies

The AES generator is the key part of our proposed AEA framework. For the AES generator, the deep feature loss and the composition of adversarial examples play a crucial role. Therefore, we analyze the necessity of two components in the ablation experiments.

Without loss of generality, we conduct experiments on SOD models to verify the effectiveness of the above components. Firstly, the original image is altered either using the multiplicative perturbation (AEA_C) or the additive perturbation (AEA_F). Secondly, only the BCE loss is used while the deep feature loss is not utilized (AEA_{ndf}) to optimize the final adversarial example. The attack performance of different ablation attack methods on different SOD models are presented in Table 4.

By comparing the complete AEA with the ablation models AEA_C and AEA_F, we can find that the composition of the AES generator has a significant impact on the attack effectiveness. Using only multiplicative perturbations or only additive perturbations alone cannot achieve satisfactory results. However, the attack method using only additive perturbations performs better than the one using only multiplicative perturbations. This is due to the multiplicative perturbations focus on changing the luminance of key pixels in the image, while additive perturbations can alter the overall pixel variations. Simultaneously using both types of perturbations allows modifications in both luminance and color, resulting in an improved attack effectiveness. By comparing the complete AEA with AEA_{ndf}, we can find that the AEA_{ndf} is almost ineffective. Adversarial examples generated without the supervision of deep feature loss lack diversity in features. This confirms the importance of deep feature loss in the AEA framework and also illustrates the higher-transferability of AES. In order to observe the attack effects of different ablation methods more intuitively, we select the EGNet as the target model without loss of generality. The prediction results are shown in Fig. 8. We can see that the prediction results of ablation models are significantly worse than those of the complete AEA.

5. Conclusion

This paper has presented a black-box adversarial attack method with high transferability, named adversarial example augmentation (AEA) framework, for the BIS task. The core idea is to generate an adversarial example set (AES) which contains a set of distinct adversarial examples

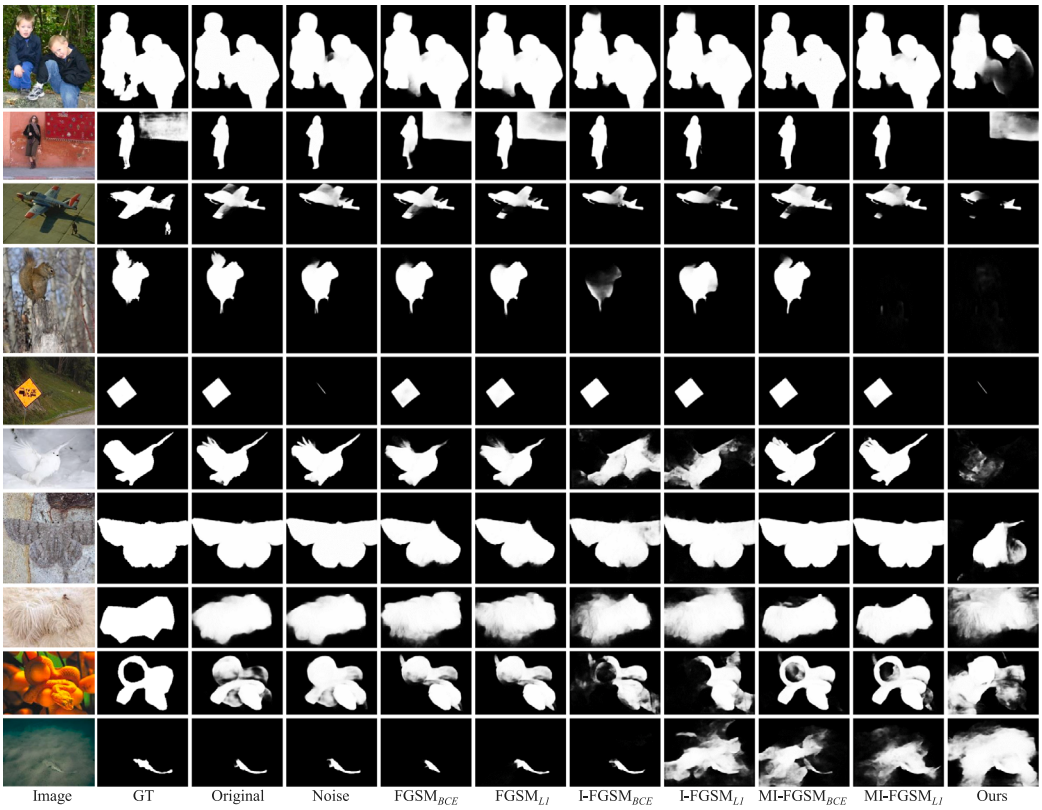

Fig. 7. Qualitative comparison of different attack methods. The first five rows are prediction results of BASNet for adversarial examples generated by different attack methods on the samples selected from ECSSD [41] dataset, and the last five rows are prediction results of FAPNet for adversarial examples generated by different attack methods on the samples selected from the COD10K [11] and CAMO [42] datasets.

Table 4
Performance metrics of the ablation attack models on the ECSSD dataset. the best results are highlighted in **Bold**.

Ablation models	PSGLoss [44]				BASNet [45]				EGNet [46]				BCMNet [47]			
	$AP\downarrow$	$M\uparrow$	$S_\alpha\downarrow$	$F_\beta^\omega\downarrow$	$AP\downarrow$	$M\uparrow$	$S_\alpha\downarrow$	$F_\beta^\omega\downarrow$	$AP\downarrow$	$M\uparrow$	$S_\alpha\downarrow$	$F_\beta^\omega\downarrow$	$AP\downarrow$	$M\uparrow$	$S_\alpha\downarrow$	$F_\beta^\omega\downarrow$
Original	0.926	0.113	0.719	0.688	0.913	0.056	0.87	0.839	0.936	0.037	0.925	0.904	0.926	0.099	0.782	0.737
AEA$_F$	0.507	0.251	0.538	0.414	0.874	0.081	0.813	0.756	**0.861**	0.071	0.861	0.799	**0.791**	0.142	0.680	0.588
AEA$_C$	0.681	0.177	0.626	0.530	0.891	0.072	0.834	0.788	0.905	0.052	0.897	0.859	0.842	0.123	0.725	0.653
AEA$_{ndf}$	0.878	0.118	0.715	0.671	0.907	0.057	0.868	0.835	0.924	0.039	0.921	0.896	0.893	0.101	0.777	0.729
AEA	**0.425**	**0.307**	**0.472**	**0.341**	**0.867**	**0.087**	**0.799**	**0.737**	0.874	**0.079**	**0.845**	**0.781**	**0.791**	**0.149**	**0.663**	**0.563**

by simultaneously optimizing the BCE loss between the prediction of the surrogate model and the pseudo-label and the deep feature loss among different adversarial examples. Given the diversity of the generated adversarial examples in the AES of the surrogate model, the optimal adversarial example for a certain target model is likely contained in our generated AES. Thus, the generated AES is expected to have high transferability. Finally, the query method is adopted to find the optimal adversarial example of a specific target model in the AES. Experimental results showcase the superiority of the proposed AEA framework for black-box attack in two representative BIS tasks, including salient object detection and camouflage object detection. However, our method may have cases where transferability is unstable because there is no guarantee that adversarial examples will necessarily fall within the wrong region of the target model. We will further investigate the transferability of adversarial examples by considering the common and unique features extracted from images by different models. In addition, our attack method also has the possibility to extend to other domains. Examples are attacks on video and audio–video object segmentation or quality assessment. Because our method essentially achieves the augmentation effect by amplifying the difference of features between adversarial examples. So, it might still be useful for more complex tasks.

CRediT authorship contribution statement

Xuebiao Zhu: Conceptualization, Methodology, Software, Writing – original draft. **Wu Chen:** Methodology, Formal analysis, Writing – review & editing. **Qiuping Jiang:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

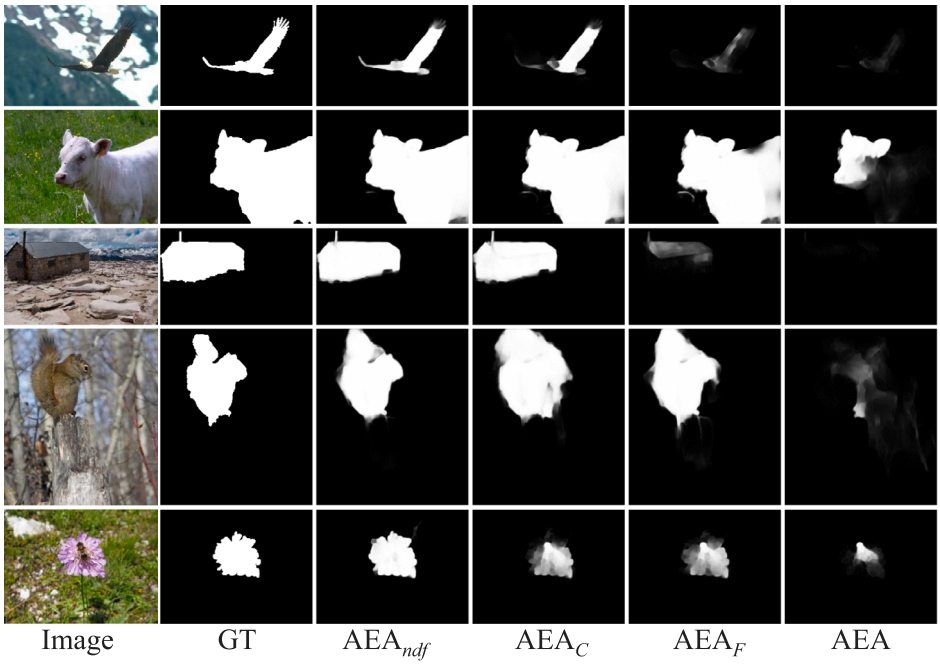

Fig. 8. Prediction results of EGNet attacked by different ablation models.

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